Fixed-difference fast-Kummer walk

KUMMER-DIFFERENTIAL-WALK-20261022

Fixed pair: K(D), K(D+G); G is the signed source generator.

Transitions D→2D or D→2D+G use three fast xADD calls.

All 1,260 targets: 3,780/3,780 exact quotient and signed-label

transitions, zero wrong or inapplicable cases.

Canonical-ratio partition: branch 0 = 1,860; branch 1 = 1,920.

It uses one field inversion per transition.

Preliminary Legendre-product partition: 2 vs 3,778; raw retained.

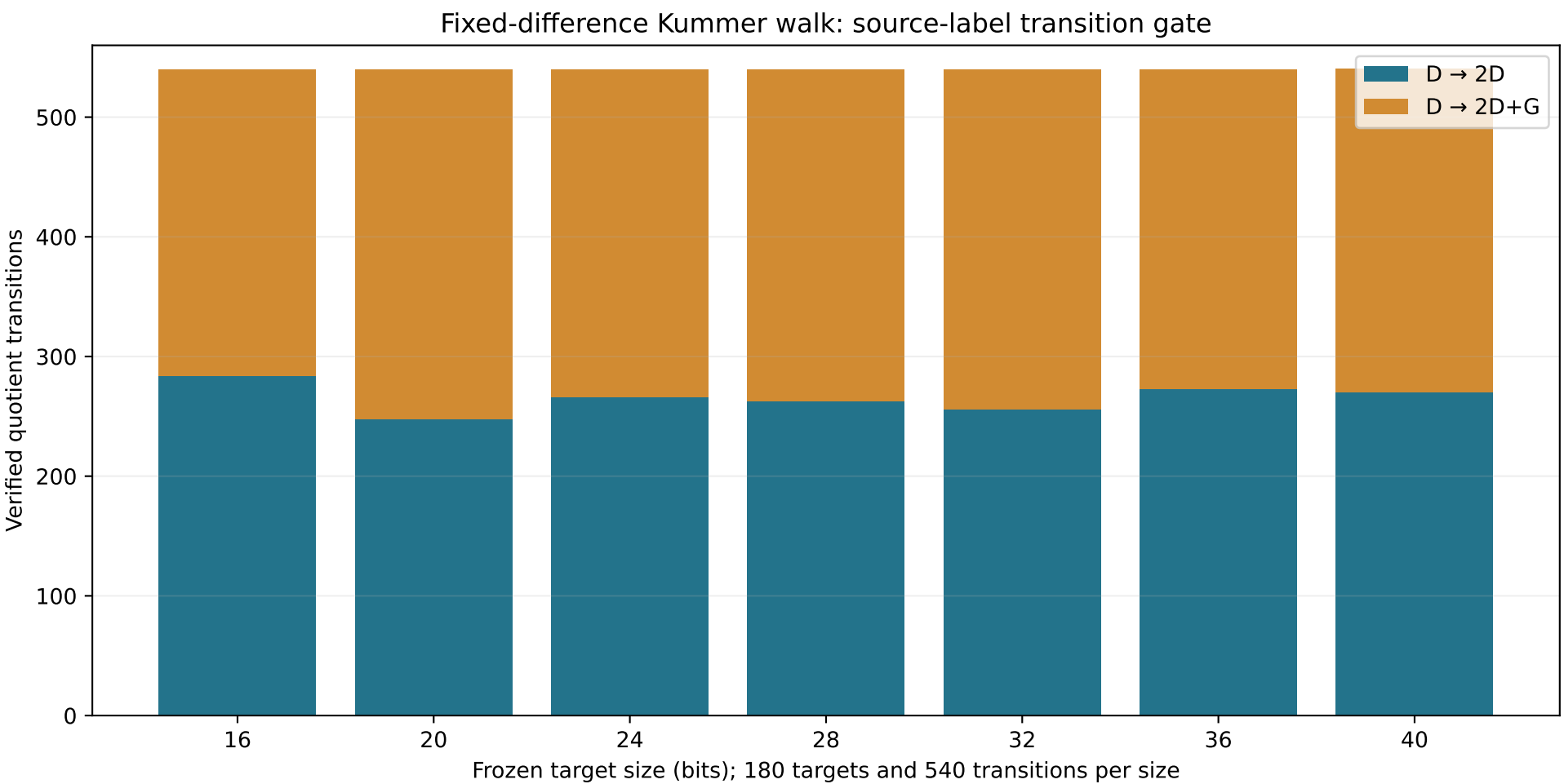
Signed relation extraction, collector and rho comparison: not run.

Candidate cost, rho cost and ratio: null.

Rows: results/transition-*.jsonl and legendre_diagnostic/.

Primary formula source: Chung–Costello–Smith, §7.1:

https://www.microsoft.com/en-us/research/wp-content/uploads/2016/02/recovery.pdf


Fixed-difference Kummer walk: source-label transition gate
D → 2D
D → 2D+G
Verified quotient transitions
0
100
200
300
400
500
16
20
24
28
32
36
40
Frozen target size (bits); 180 targets and 540 transitions per size